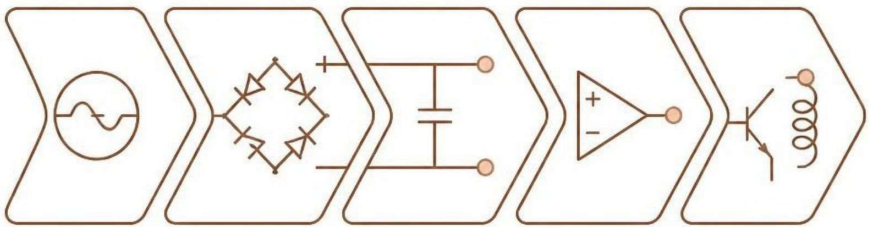
Static Instantaneous (50): Block Diagram



Instantaneous Element Logic

The static 50 element is considerably simpler than the 51 — it contains no timing network. The key signal path is:

Current → Rectifier → Level Comparator → Instantaneous Trip Output

- No RC timing circuit — comparison is direct and immediate
- Pickup threshold set by an adjustable reference voltage
- Operating time is typically **8–16 ms** (half to one cycle)
- Output transistor drives an interposing output relay to trip the breaker